3) Consider the unnormalized relation R with six attributes ABCDEF and the following functional dependencies:

AB → CDE

B → F

C → D

a) What is the key(s) for the relation?

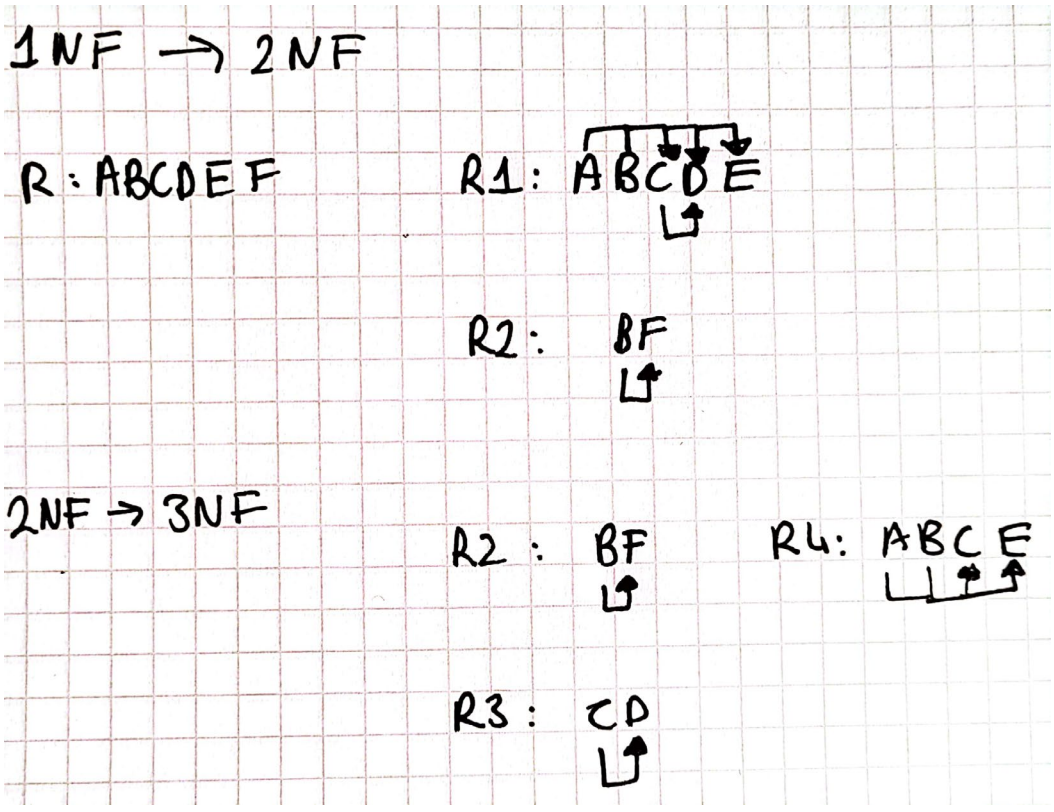
b) What is the normal form of this relation? Explain it.

c) Decompose R into 3NF relations step by step if it is not in 3NF.

a) Composite Key = (A, B)

b) 1NF because there is no multi valued attributes.

c)



4) Consider the following normalized relations from a database in a large retail chain:

STORE (StoreID, Region, ManagerID, SquareFeet)

EMPLOYEE (EmployeeID, WhereWork, EmployeeName, EmployeeAddress)

DEPARTMENT (DepartmentID, ManagerID, SalesGoal)

SCHEDULE (DepartmentID, EmployeeID, Date)

What opportunities might exist for denormalizing these relations when defining the physical records for this database? Under what circumstances would you consider creating such denormalized records?

STORE
Store Name
Region
Manager Name
Square Feet

EMPLOYEE
Name
Working Store

Address

DEPARTMENT
Department Name
Manager Name
Sales Goal

SCHEDULE
Department Name
Employee Name
Date

5) Consider the following two relations for Millennium College:

STUDENT (StudentID, StudentName, CampusAddress, GPA)

REGISTRATION (StudentID, CourseID, Grade)

Following is a typical query against these relations:

```
SELECT Student.StudentID, StudentName, CourseID, Grade
FROM Student, Registration
WHERE Student.StudentID = Registration.StudentID AND GPA > 3.0
ORDER BY StudentName;
```

a) On what attributes should indexes be defined to speed up this query? Give the reasons for each attribute selected.

b) Write SQL commands to create indexes for each attribute you identified in part a.

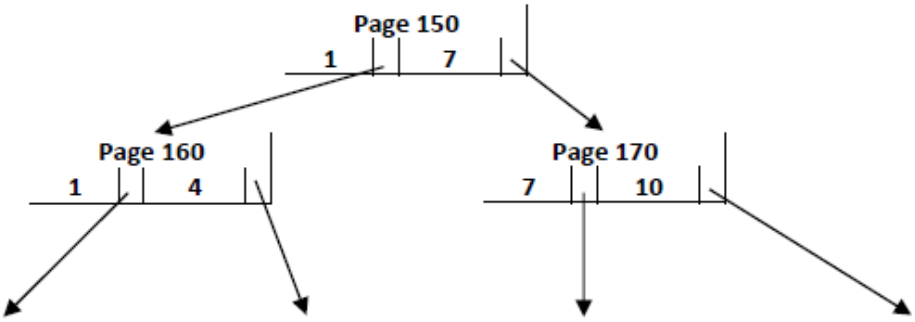
a) StudentID in Registration and Student should be indexed because we do inner join with Student table. Indexing StudentID will make finding registration record of student faster. We should index GPA because we search GPAs greater than 3.0 every time. Also we should create index for StudentName in Student table because indexes create an ordered structure so we won't need to order students again and again.

b)
```
CREATE INDEX st_name
ON Student (StudentName)
CREATE INDEX st_gpa
ON Student (GPA)
CREATE INDEX st_id
ON Registration (StudentID)
CREATE INDEX st_id2
ON Student (StudentID)
```

6) You have a STUDENT table that has SID, Name, and Age columns. Which data pages are accessed to execute the queries below, under situations given at (a) and (b)? (Assume that index seek is used whenever possible)

STUDENT		
SID	**Name**	**Age**

a) The table has a *clustered index* on **SID** column, and no other indexes. The index structure and data is stored on data pages as the following:



Page 110				Page 120				Page 130				Page 140			
01	1	Ayşe	2	**01**	4	Adem	4	**01**	7	Mert	8	**01**	10	Ceren	13
02	2	Fatma	7	**02**	5	Mehtap	4	**02**	8	Mehmet	10	**02**	11	Ali	16
03	3	Can	11	**03**	6	Ahmet	6	**03**	9	Zeynep	17	**03**	12	Yavuz	16

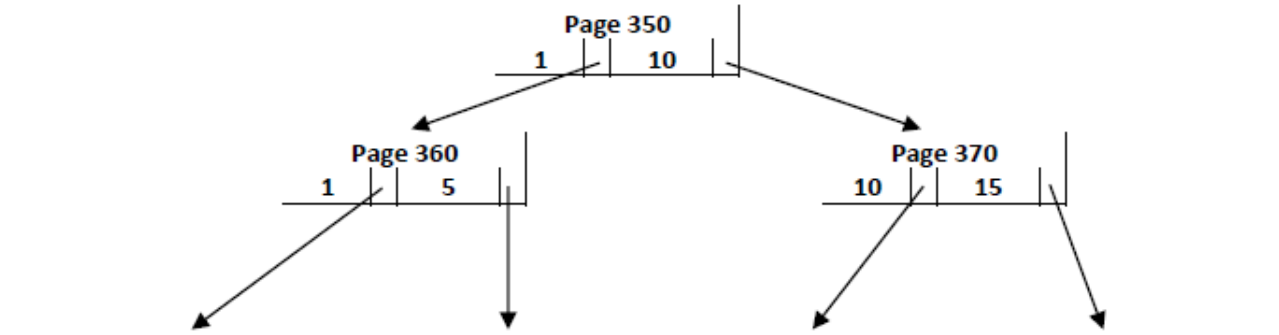
i) **Query 1:** select Name from STUDENT where SID < 11

ii) **Query 2:** select * from STUDENT where Age = 16

iii) **Query 3:** select * from STUDENT where SID = 7

a. i) Page 110, 120, 130, 140
ii) Page 140
iii) Page 130

b) The table has a *non-clustered index* on **Age** column, and no other indexes. The index structure and data is stored on data pages as the following:



Page 220		Page 230		Page 240		Page 250	
2	100:01	**6**	110:03	**10**	120:02	**16**	130:02
4	110:01	**7**	100:02	**11**	100:03	**16**	130:03
4	110:02	**8**	120:01	**13**	130:01	**17**	120:03

Page 100				Page 110				Page 120				Page 130			
01	1	Ayşe	2	**01**	4	Adem	4	**01**	7	Mert	8	**01**	10	Ceren	13
02	2	Fatma	7	**02**	5	Mehtap	4	**02**	8	Mehmet	10	**02**	11	Ali	16
03	3	Can	11	**03**	6	Ahmet	6	**03**	9	Zeynep	17	**03**	12	Yavuz	16

i) **Query 1:** select Age, Name from STUDENT where SID < 9

ii) **Query 2:** select Age from STUDENT where Age < 8

iii) **Query 3:** select * from STUDENT where Age = 8

d) i) All pages will be accessed
ii) Pages from 100 to 120 will be accessed (100,101,102…120)
iii) Page 120 will be accessed